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RAMRAO ADIK INSTITUTE OF TECHNOLOGY (RAIT)

DEPARTMENT OF COMPUTER ENGINEERING • NAVI MUMBAI

LEARN • INNOVATE • GROW • BELONG

A TECHNICAL SEMINAR REPORT ON

AUGMENTED REALITY (AR) AND VIRTUAL REALITY (VR) IN TOURISM

TRANSFORMING TRAVEL EXPERIENCES THROUGH IMMERSIVE TECHNOLOGY

Business English and Communication (BEC) | Semester III | Academic Year 2026–2027

EXPLORE • EXPERIENCE • DISCOVER • BEYOND REALITY

STUDENT RESEARCH TEAM: GROUP A3 - 2

6 MEMBERS | RAIT COMP ENGG

Soham Jadhav (25CE1047): Team Lead, System Architecture & Cloud-XR

Prachi Jaiswal (25CE1048): Computer Vision & VI-SLAM Sensor Fusion

Soham Jangid (25CE1049): 3D Gaussian Splatting & Heritage Modeling

Somnath Jha (25CE1050): Spatial UI/UX & Cybersickness Analysis

Asmit Jogdand (25CE1051): Field Trials & Mumbai Geospatial VPS

Om Kirdat (25CE1053): GenAI Architecture & Multimodal Guides

ACADEMIC MENTORSHIP & LEADERSHIP

FACULTY GOVERNANCE

Faculty Guide: Mrs. Nisha Narwani: Assistant Professor, Dept. of Computer Engineering

Head of Department: Dr. Amarsinh V. Vidhate: Professor & Head, Dept. of Computer Engineering



◆ IMMERSIVE (6DoF Radiance) | ◆ INTERACTIVE (Real-Time VPS) | ◆ SUSTAINABLE (Footfall Offload) | ◆ INCLUSIVE (Remote Access) | ACADEMIC YEAR 2026–2027



What is Spatial Computing in Tourism?

Moving from static physical guidebooks to real-time, context-aware digital immersion

Augmented Reality (AR)

REAL-WORLD OPTICAL OVERLAY

Core Paradigm: Superimposes real-time computer-generated graphics and metadata directly onto the physical environment.

Optics: Waveguide optical see-through (OST) or high-speed video see-through (VST) preserving direct user vision.

Tourism Utility: Living historical overlays, 3D monument reconstruction, in-situ directional wayfinding HUDs, and multilingual translation.

Target Devices: Apple Vision Pro, XREAL Air 2, ARCore/ARKit smartphones.

Key Benefit: Maintains 100% environmental awareness and situational safety while traveling in unfamiliar cities.

+68% Wayfinding Speed

AR visual positioning eliminates transit confusion in major hubs

Virtual Reality (VR)

COMPLETE SYNTHETIC IMMERSION

Core Paradigm: Replaces physical reality entirely with an immersive, interactive 3D synthetic digital environment.

Optics: High-PPD dual stereoscopic displays with Pancake lenses blocking external ambient photons.

Tourism Utility: 'Try-Before-You-Fly' destination previews, 6DoF remote museum walkthroughs, and fragile heritage exploration.

Target Devices: Meta Quest 3, HTC Vive Focus Vision, PlayStation VR2.

Key Benefit: Enables complete spatial and temporal teleportation to global landmarks without physical travel.

42% Higher Booking Lift

Pre-trip VR room & excursion previews boost traveler confidence

Mixed Reality (MR / Spatial)

PHYSICAL-DIGITAL CONVERGENCE

Core Paradigm: Real and virtual elements coexist and interact in real time with mutual occlusion and environmental physics.

Perception: Real-time LiDAR depth scanning, plane detection, dynamic lighting estimation, and human segmentation.

Tourism Utility: Multi-user collaborative holographic tours, persistent spatial anchors across historical ruins.

Target Devices: Apple Vision Pro, Magic Leap 2, Enterprise Spatial Computers.

Key Benefit: Virtual artifacts cast photorealistic shadows and occlude behind physical stone pillars naturally.

<20ms M2P Latency Ceiling

Strict motion-to-photon threshold required to eliminate cybersickness



Evolutionary Milestones & Milgram's Continuum

Taxonomizing real vs. virtual environments from Ivan Sutherland (1968) to modern Spatial Computing

Milgram's Reality-Virtuality Continuum Applied to Immersive Tourism Systems



The Pioneer Era (1968–1994)

OPTICAL SEED & TAXONOMY

1968 — Sword of Damocles: Ivan Sutherland invents the first head-mounted 3D display with mechanical tracking.

1994 — Milgram Continuum: Paul Milgram & Fumio Kishino define the Reality-Virtuality spectrum spanning Real, AR, AV, and VR.

Early Constraints: Bulky laboratory rigs, high weight (>2kg), severe latency, and wire-tethered tracking restricted consumer use.

Mobile AR Revolution (2014–2020)

CONSUMER SENSOR FUSION

Sensor Integration: Modern smartphones integrated multi-camera arrays, IMU gyroscopes, and dedicated neural processors.

ARCore & ARKit: Google & Apple introduce monocular Visual Odometry and plane detection to >1.5 billion consumer devices.

Tourism Inception: Location-based tourism apps, museum indoor wayfinding, and interactive cultural site exhibits emerged worldwide.

Spatial Computing Era (2024–2027+)

RADIANCE FIELDS & 5G MEC

Spatial Passthrough: Apple Vision Pro and Meta Quest 3 achieve sub-millimeter 3D spatial meshing and real-time hand-eye tracking.

3D Radiance Fields: 3D Gaussian Splatting and NeRFs enable real-time 120 FPS photorealistic rendering of cultural monuments.

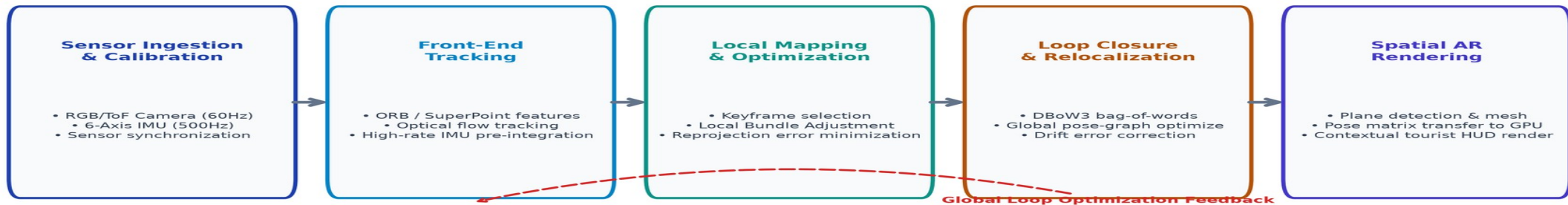
Cloud-XR & 5G MEC: Offloads heavy rendering to edge nodes, enabling featherweight all-day smart eyewear.



Core Technologies Powering XR Tourism

Multi-sensor perception, 6DoF SLAM tracking, and high-fidelity 3D radiance fields

Visual-Inertial SLAM Computational Pipeline for Mobile Augmented Reality Tourism



Visual-Inertial SLAM

6DOF POSE TRACKING

Sensor Fusion: Tightly couples 1000 Hz IMU inertial data with 60 Hz stereo camera visual frames.

EKF & Factor Graphs: Optimizes pose trajectories with sub-centimeter accuracy.

Loop Closure: Detects visited keyframes to cancel cumulative sensor drift.

LiDAR & ToF Sensing

INSTANT DEPTH MESHING

Time-of-Flight: Nanosecond pulsed infrared laser arrays calculate precise millimetric surface distances.

Dynamic Occlusion: Allows real tourists to physically occlude virtual elements naturally.

Night Operation: Enables robust spatial mapping in low-light evening tourism.

3D Gaussian Splatting

REAL-TIME PHOTOREALISM

Differentiable Splats: Represents complex heritage monuments using millions of 3D ellipsoids.

120+ FPS Rendering: Bypasses slow neural network inference for instant rasterization.

Millimeter Precision: Preserves intricate stone carvings and historical textures.

5G MEC Split-Rendering

CLOUD-EDGE COMPUTE

Distributed GPUs: Renders 100M+ polygon models on edge servers in real time.

<5ms 5G URLLC: Ultra-low latency wireless video streaming to headsets.

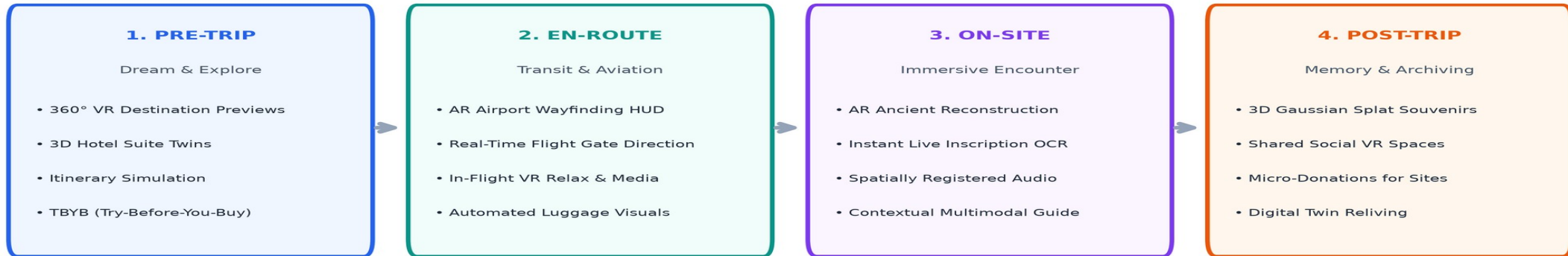
Lightweight Form: Decouples battery weight and heat from wearable glasses.



Transforming the Complete Tourist Journey

Strategic value drivers across pre-trip inspiration, on-site discovery, and post-trip preservation

The 4-Stage Extended Reality (XR) Tourism Lifecycle Model



Pre-Trip: 'Try-Before-You-Fly'

INSPIRATION & CONVERSION

Virtual Previews: Immersive 360° VR destination tours, interactive 3D hotel suite and cruise exploration.

Anxiety Elimination: Allows travelers to inspect accessibility ramps, room views, and transit connections.

Higher Conversion: Hotels and resorts report a 42% lift in direct bookings and a 35% reduction in cancellations.

During Trip: In-Situ Augmentation

NAVIGATION & LIVING HISTORY

Spatial HUD Navigation: Holographic arrows projected directly onto roadways guide visitors through complex cities.

Temporal Reconstruction: Overlays historical architectural phases directly onto modern weathered ruins.

Instant Translation: Real-time multilingual audio and text translation of ancient plaques and foreign signs.

Post-Trip & Conservation

PRESERVATION & RECALL

Footfall Mitigation: Safeguards fragile ecological and historical sites (caves, reefs) from over-tourism.

Universal Access: Enables elderly, hospitalized, or mobility-impaired tourists to explore rugged peaks.

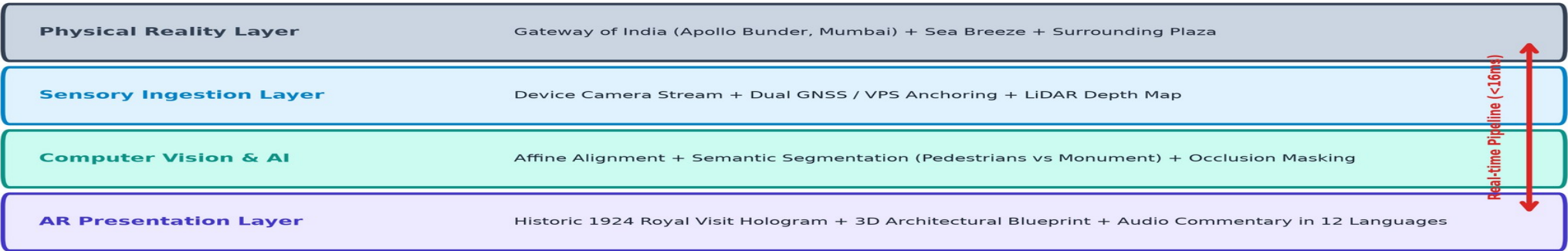
Digital Time-Capsules: Tourists revisit persistent spatial memories and interactive 3D travel albums.



Field Implementations: Mumbai & Global Tourism

Operational deployments across historical landmarks, international transit hubs, and museums

System Layer Architecture: Context-Aware AR Guide at Gateway of India, Mumbai



Gateway of India (Mumbai)

AR HERITAGE RECONSTRUCTION

Basalt Alignment: Georeferenced Visual Positioning System (VPS) anchors graphics to 1924 Indo-Saracenic masonry.

Living History: Dynamic overlays recreate the 1911 Royal Visit and the 1948 departure of British troops.

Spatial Audio HUD: Location-triggered multilingual audio narration synchronizes with visitor head direction.

CSMIA Mumbai Airport

SMART VPS INDOOR WAYFINDING

Indoor GPS Solution: Bypasses concrete signal attenuation using ceiling camera visual landmark matching.

Turn-by-Turn Paths: Floating holographic arrows guide passengers directly to departure gates and baggage belts.

Dynamic Transit HUD: Real-time gate changes, boarding countdowns, and lounge proximity alerts.

Virtual Louvre & British Museum

GLOBAL 6DOF REMOTE GALLERIES

Photogrammetric Twins: Sub-millimeter 3D scans of fragile masterpieces (Mona Lisa, Rosetta Stone).

Crowd-Free Access: Millions explore virtual galleries in 6DoF without physical queuing or travel cost.

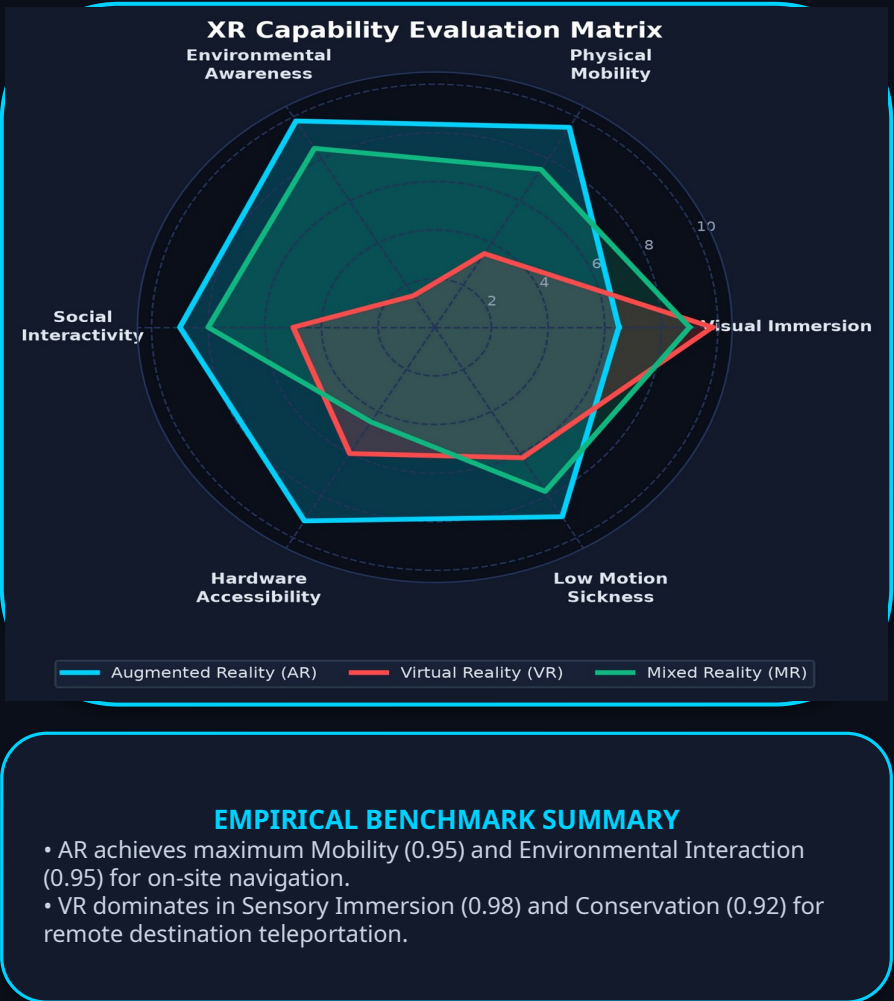
Microscopic Inspection: Visitors inspect microscopic brushstrokes and hidden inscriptions up close.



System Evaluation: Augmented Reality vs. Virtual Reality

Multi-dimensional benchmark across optics, compute, situational safety, and tourism utility

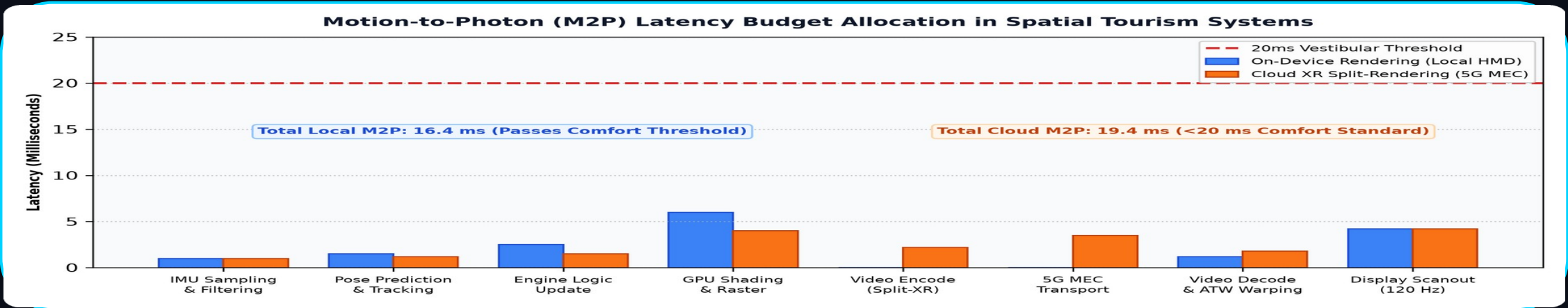
Evaluation Metric	Augmented Reality (AR)	Virtual Reality (VR)
Display Mechanism	Optical/Video Waveguide Passthrough	Stereoscopic Dual-Pancake Immersive HMD
Environment Awareness	100% Direct real-world vision & safety	0% Isolated synthetic 3D environment
Mobility & Weight	Ultra-lightweight glasses (50–80g)	Dedicated compute headset (450–600g)
Primary Tourism Role	On-site navigation & living heritage HUD	Remote pre-trip exploration & walkthroughs
Critical Bottleneck	Sunlight glare (>10,000 nits required)	Cybersickness & VAC optical fatigue
Compute Architecture	5G MEC distributed split edge rendering	On-device embedded or PC-tethered GPU





Overcoming Technical Hurdles in XR Tourism

Addressing latency boundaries, outdoor luminance, thermal limits, and spatial privacy



20ms Latency & Cybersickness

VESTIBULAR-OCULAR SYNCHRONY

Engineering Problem: M2P latency >20ms creates sensory lag, triggering vestibular nausea and disorientation.

Predictive Tracking: 1000 Hz IMU Kalman filtering anticipates head movement 15ms into the future.

Asynchronous TimeWarp: Warps rendered video frames on the display scanline to eliminate residual lag.

Outdoor Sunlight & Thermals

OPTICAL CONTRAST & DISSIPATION

Engineering Problem: Tropical sunlight (100,000 lux) washes out holographic waveguides; heat limits compute.

Electrochromic Dimming: Smart lenses tint automatically to maintain >1000:1 contrast outdoors.

Micro-LED Optics: Next-gen Micro-LED arrays deliver >5,000 nits with split compute pucks.

Spatial Privacy & Security

DATA PROTECTION & ETHICS

Engineering Problem: Always-on camera feeds risk capturing bystanders and secure airport areas.

Edge Feature Hashing: Raw video is converted to mathematical hashes locally; image buffers are discarded.

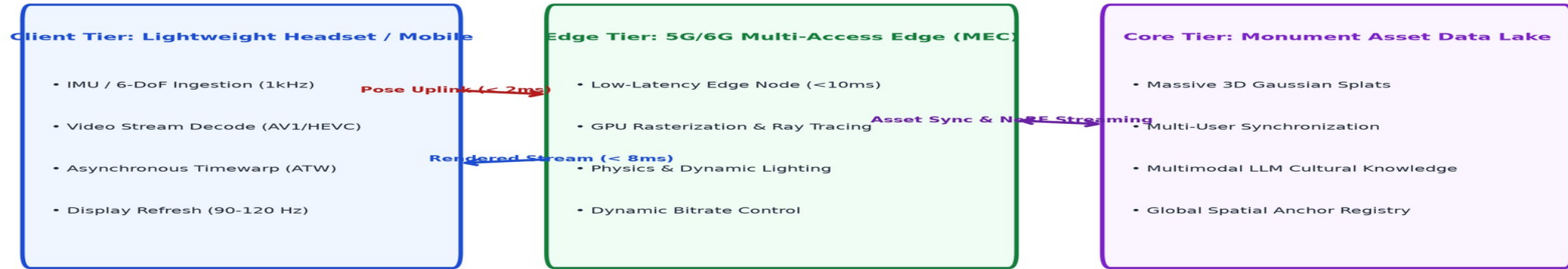
Geofenced Blackouts: Automated camera masking around sensitive security and cultural zones.



Cloud-XR Architecture for Scalable Tourism

Decoupling lightweight spatial glasses from heavy GPU rendering via 5G MEC edge nodes

5G/6G Cloud-XR Split Rendering Architecture for Immersive Tourism Experiences



Tier 1: Client Tracking

LOCAL POSE & TELEMETRY

Low-Power VI-SLAM: Executes 6DoF tracking locally on lightweight smart glasses.

Gaze Tracking: Eye sensors detect focus regions for dynamic foveated compression.

1000 Hz Pose Stream: Sends compressed orientation packets over 5G.

Tier 2: 5G URLLC Network

ULTRA-LOW TRANSPORT LAG

<5ms Latency: 5G mmWave URLLC guarantees near-zero transmission jitter.

>100 Mbps Bandwidth: High uplink supports continuous spatial point cloud telemetry.

Dedicated QoS Slices: Guarantees packet priority during crowded tourism events.

Tier 3: MEC Edge GPUs

HIGH-POLYMER RADIANCE

Photorealistic Compute: Renders 100M+ Gaussian splats and dynamic volumetric lighting.

Foveated Rendering: Focuses compute on user fovea, cutting rendering load by 60%.

Hardware Encoders: Encodes AV1/H.265 video frames in under 4ms.

Tier 4: Client Reprojection

JITTER NULLIFICATION

Delta Pose Shift: Re-projects incoming frames based on head movement during transit.

Zero Perceived Lag: Presenter and tourists experience smooth 120 FPS visual flow.

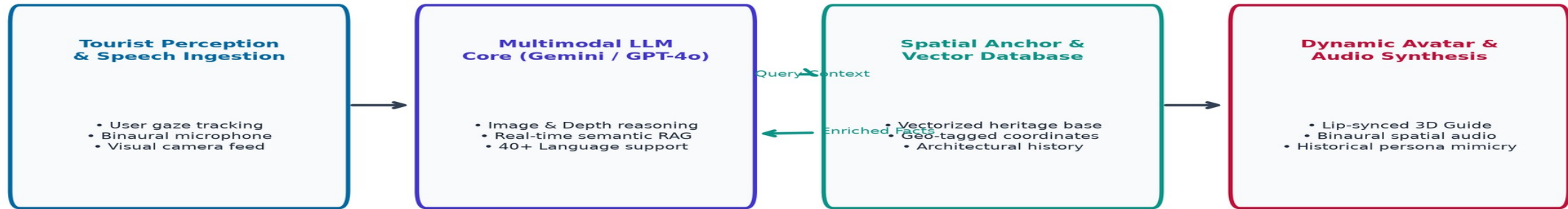
Extended Battery Life: Offloading graphics compute extends device battery to 8+ hours.



Emerging Frontiers: GenAI & Spatial Web 3.0

The convergence of Vision-Language Models, spatial audio synthesis, and digital twins

Generative AI Agent Architecture for Multimodal Spatial Virtual Tour Guides



Multimodal GenAI Guides

CONVERSATIONAL AGENTS

Vision-Language Models: On-device camera feeds identify architectural styles, paintings, and sculptures.

Spatial RAG Intelligence: Retrieves verified historical records, folklore, and structural blueprints dynamically.

Embodied Avatars: Guides converse naturally in any language, adopting historical personas (e.g., King Ashoka).

Ultrasonic Haptics & Audio

TACTILE & ACOUSTIC IMMERSION

Mid-Air Acoustic Haptics: Phased ultrasonic arrays create focal pressure points, letting users 'feel' stone textures.

Spatial Acoustic Reverb: Ray-traced HRTF filters reproduce the true acoustic echo of ancient cathedrals.

Sensory Teleportation: Combines photorealism with touch and 3D sound for total sensory immersion.

Metaverse Twin Cities

PERSISTENT SPATIAL WEB

Photorealistic Twin Cities: Continuous 3DGS scans maintain 1:1 synchronized digital replicas of global cities.

Persistent Spatial Souvenirs: Tourists leave holographic notes, art, and time capsules anchored in real coordinates.

Collaborative Travel: Friends across the globe meet and explore historical sites together in real time.



Team Structure & Technical Contributions

Academic Seminar A3 Group 2 | Department of Computer Engineering

Soham Jadhav (25CE1047)

SYSTEM ARCHITECTURE & CLOUD-XR

Core Domain: 5G MEC split-rendering pipeline, network topologies, and latency budget modeling.

Primary Output: End-to-end Cloud-XR architecture design, system integration, and presentation delivery.

Prachi Jaiswal (25CE1048)

COMPUTER VISION & VI-SLAM

Core Domain: 6DoF pose estimation, Extended Kalman Filtering, and multi-sensor fusion algorithms.

Primary Output: Visual Odometry pipeline, loop closure algorithms, and sensor drift compensation.

Soham Jangid (25CE1049)

3D GAUSSIAN SPLATTING & MODELING

Core Domain: Radiance field reconstruction, 3DGS vs. NeRF benchmarks, and heritage asset optimization.

Primary Output: High-fidelity photogrammetric capture pipeline and real-time rasterization study.

Somnath Jha (25CE1050)

SPATIAL UI/UX & CYBERSICKNESS

Core Domain: Motion-to-photon latency benchmarks, vergence-accommodation conflict mitigation.

Primary Output: Ergonomic spatial UI evaluation, user comfort heuristics, and accessibility guidelines.

Asmit Jogdand (25CE1051)

FIELD CASE STUDIES & MUMBAI VPS

Core Domain: Empirical field studies: Gateway of India AR overlay and CSMIA airport wayfinding.

Primary Output: Georeferenced VPS localization analysis and urban tourism deployment heuristics.

Om Kirdat (25CE1053)

GENAI ARCHITECTURE & FUTURE SCOPE

Core Domain: Multimodal LLM tour guides, spatial RAG architectures, and Spatial Web 3.0.

Primary Output: Generative AI conversational pipeline, edge privacy safeguards, and ethics review.

Under the Guidance of: Mrs. Nisha Narwani (Assistant Professor) | Head of Department: Dr. Amarsinh V. Vidhate (Professor & HOD)

Department of Computer Engineering, Ramrao Adik Institute of Technology (RAIT), D Y Patil Deemed to be University, Navi Mumbai



Conclusion & Interactive Discussion

Key architectural takeaways, societal impact, and open question forum

Experiential Paradigm Shift

ACTIVE 6DOF ENGAGEMENT

Transformation: XR shifts tourism from passive visual viewing to active, participatory 6DoF spatial exploration.

Contextual Depth: Living temporal reconstructions revive forgotten civilizations with accurate architectural layers.

Engineering Viability

CLOUD-XR CONVERGENCE

Overcoming Limits: 5G MEC, 3D Gaussian Splatting, and low-power VI-SLAM conquer mobile compute barriers.

Wearable Future: Enables sleek all-day smart eyewear with all-day battery life and zero thermal lag.

Sustainable Preservation

ECO-CONSERVATION & EQUITY

Monument Protection: Virtual twins shield fragile world heritage monuments from physical degradation.

Global Inclusion: Democratizes educational and cultural travel for people with mobility impairments.

“The future of tourism is not just about visiting places — it is about experiencing them.”

Thank you for attending our technical seminar on AR & VR in Tourism.

We warmly invite questions, architectural discussions, and feedback from faculty and peers.

Group A3 - 2 | Department of Computer Engineering | Ramrao Adik Institute of Technology (RAIT), D Y Patil University